

CodeVS

Information System Design Laboratory

Project Report

CSE 3120

Scrum Master	Rafsan Riasat Roll: 2207006
Developer (Backend / Game Engine)	Utsa Roy Roll: 2207027
Developer (Frontend / UI)	Adib Raian Roll: 2207020

Platform: Web Application (React + Firebase)

Methodology: Agile Scrum | 3 Sprints

Version Control: Git / GitHub (repository: narukami00/CodeVS)

Project Management: Jira Cloud (codevs-team.atlassian.net)

1. Introduction & Project Overview

CodeVS is a real-time multiplayer web application designed to make competitive programming practice engaging and social. Two players are matched together and simultaneously type out a randomly selected code snippet in a chosen programming language. Performance is measured in Words Per Minute (WPM) and Accuracy (%), with results persisted to a global leaderboard.

Technology Stack

Component	Technology / Tool
Frontend Framework	React (Vite build tool)
Styling	CSS / Tailwind utilities, custom hacker-themed dark design
Backend / Database	Google Firebase — Firestore (documents) + Realtime Database (presence & progress)
Authentication	Firebase Authentication (Email / Password)
Hosting	Firebase Hosting
Project Management	Jira Cloud (Scrum board, Epics, Sprints, Sub-tasks)
Version Control	Git + GitHub, with Jira integration for branch naming
Test Management	Zephyr Scale for Jira

Project Objectives

- Deliver a fully functional real-time multiplayer typing game with sub-second Firebase synchronisation.
- Support six programming languages: C, Python, JavaScript, PHP, C++, and Java.
- Implement a persistent global leaderboard ranking players by WPM and accuracy.
- Maintain a clean Agile Scrum process across three well-defined sprints.
- Ensure full mobile responsiveness for the application.

2. Core Features

User Authentication

Secure email and password login/registration powered by Firebase Authentication. Authenticated sessions persist across browser refreshes.

Quick Match & Custom Rooms

Players can enter a global matchmaking queue for a quick match or create a private room with a unique 6-character Room ID to invite a specific opponent.

Language Selection

Players may select their preferred programming language (C, Python, JavaScript, PHP, C++, Java). The matchmaking system pairs players who share a language preference and draws a snippet from the appropriate pool.

Real-Time Typing Race

Both players type the same code snippet simultaneously. Character-level correctness is validated in real-time. The opponent's progress is tracked via Firebase listeners and shown as a live progress bar.

Performance Metrics

On completion, the system computes WPM (Words Per Minute), Accuracy (%), and ranks the winner.

Result Screen

A dedicated animated result screen reveals the winner, both players' scores, and provides options to initiate a Rematch or return to Home.

Rematch Logic

Either player can request a rematch after a game. Both must accept; if agreed, the session resets with a fresh snippet. Decline gracefully navigates back to the home screen.

Global Leaderboard

A real-time ranked leaderboard persists the top WPM scores across all sessions, publicly accessible.

Mobile Responsiveness

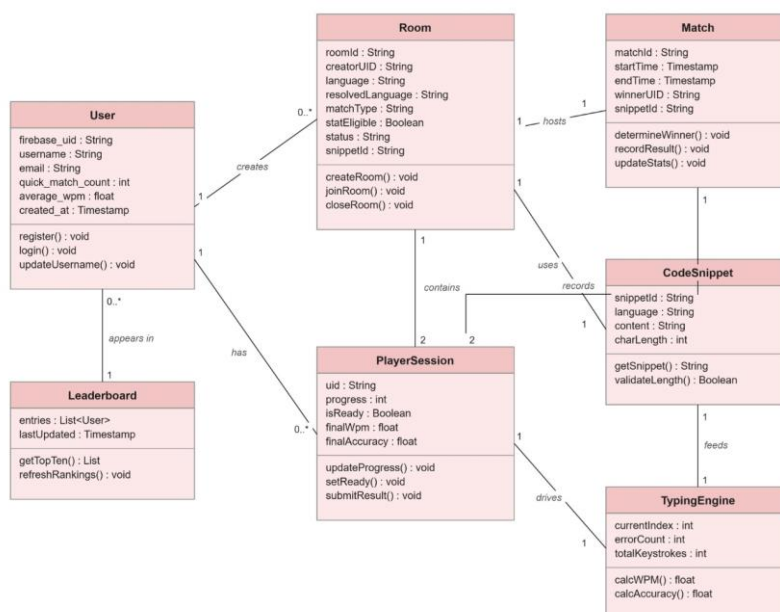
The application adapts to mobile viewports with touch-optimised controls and simplified layouts.

3. System Architecture & Design

This section presents the key structural and behavioural diagrams that define the architecture of CodeVS. Diagrams are provided externally by the Scrum Master and Developers; placeholder sections below indicate where each diagram should be inserted. Screenshot placeholders are also noted where Jira or GitHub screenshots are recommended.

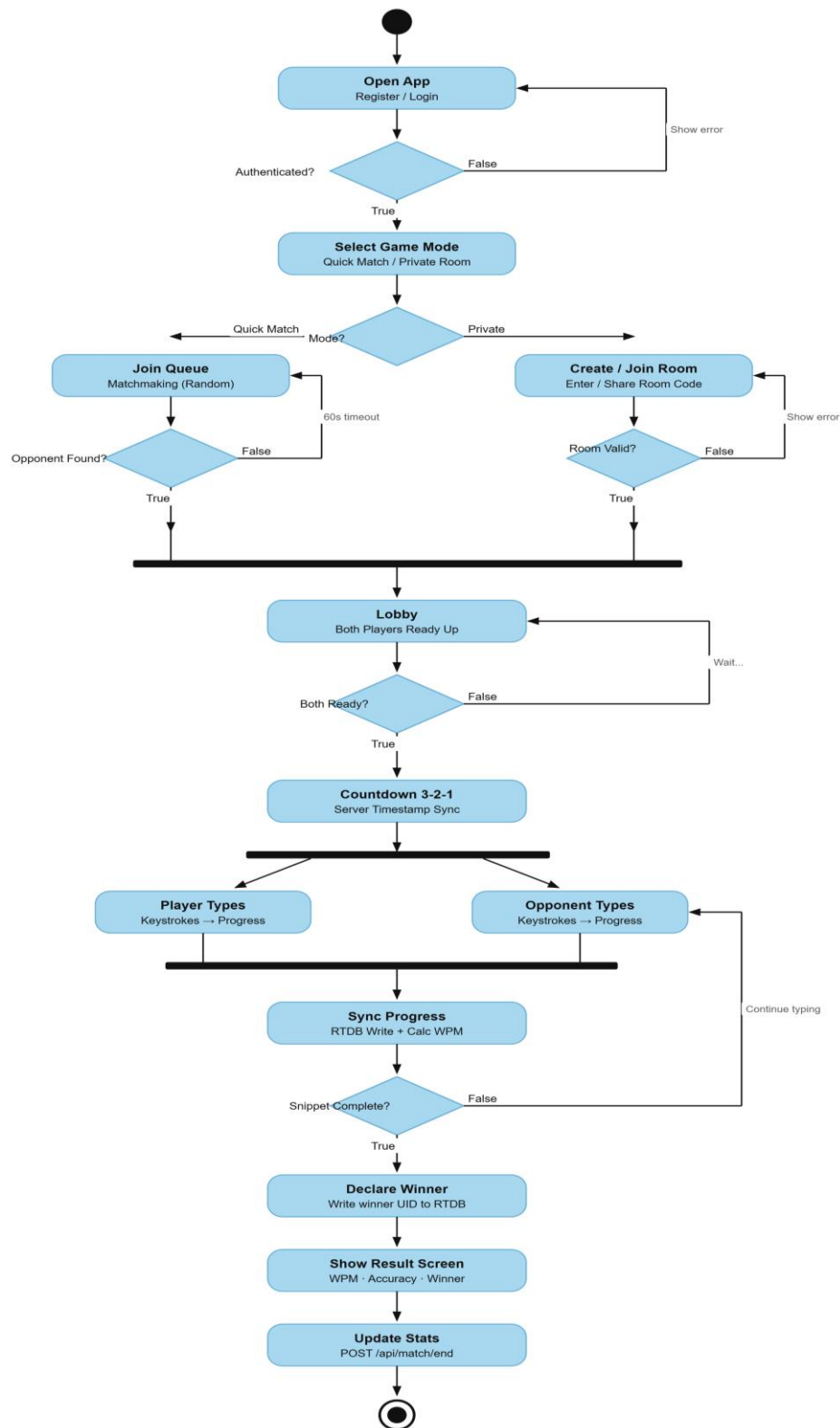
3.1 UML Class Diagram

Illustrates the major classes (User, Room, Game, Snippet, Leaderboard) and their associations, multiplicities and methods as represented in the React component and Firebase data model.



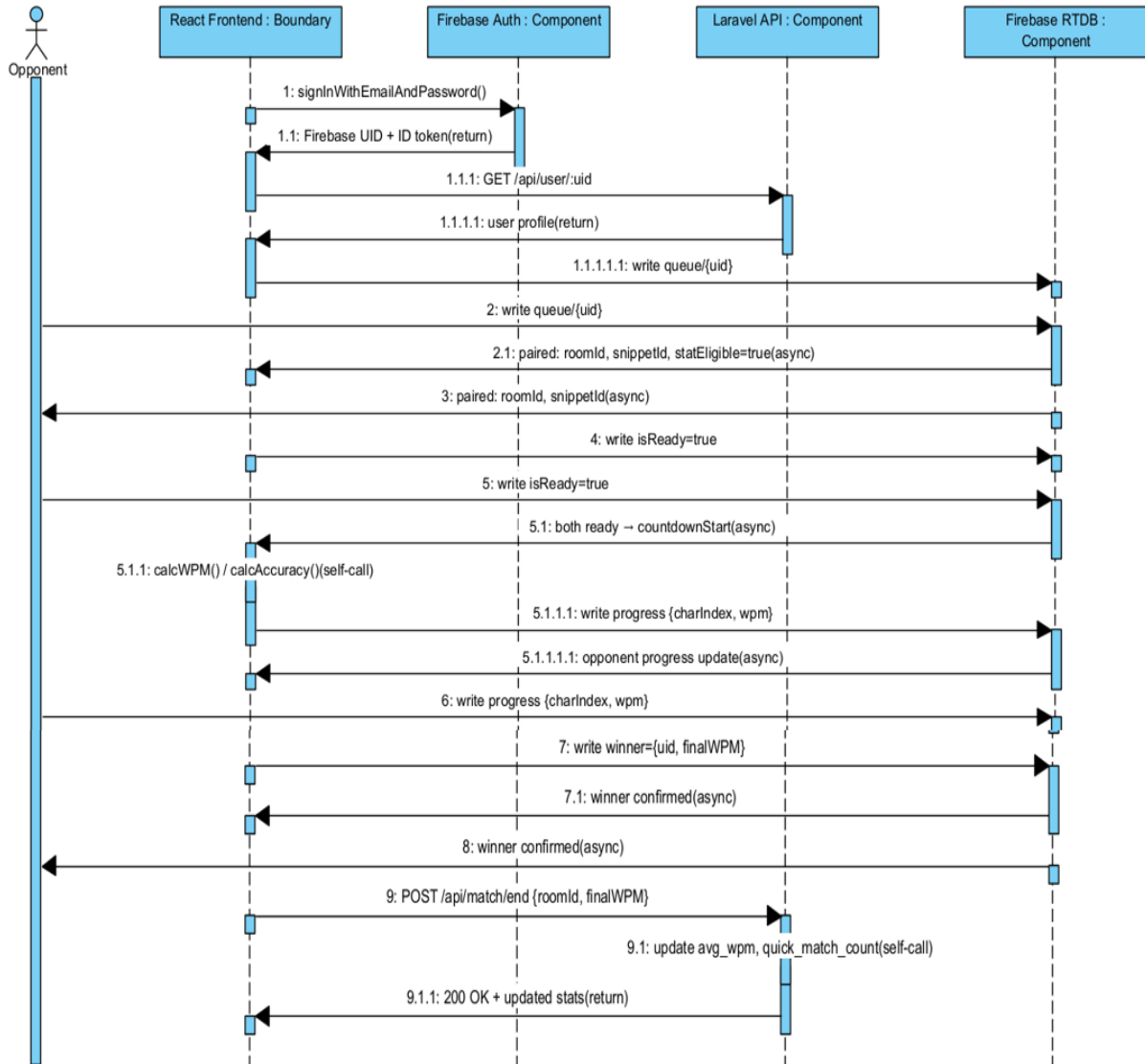
3.2 Activity Diagram

Depicts the full user activity flow from authentication through matchmaking, the live game loop, and the post-match result sequence.



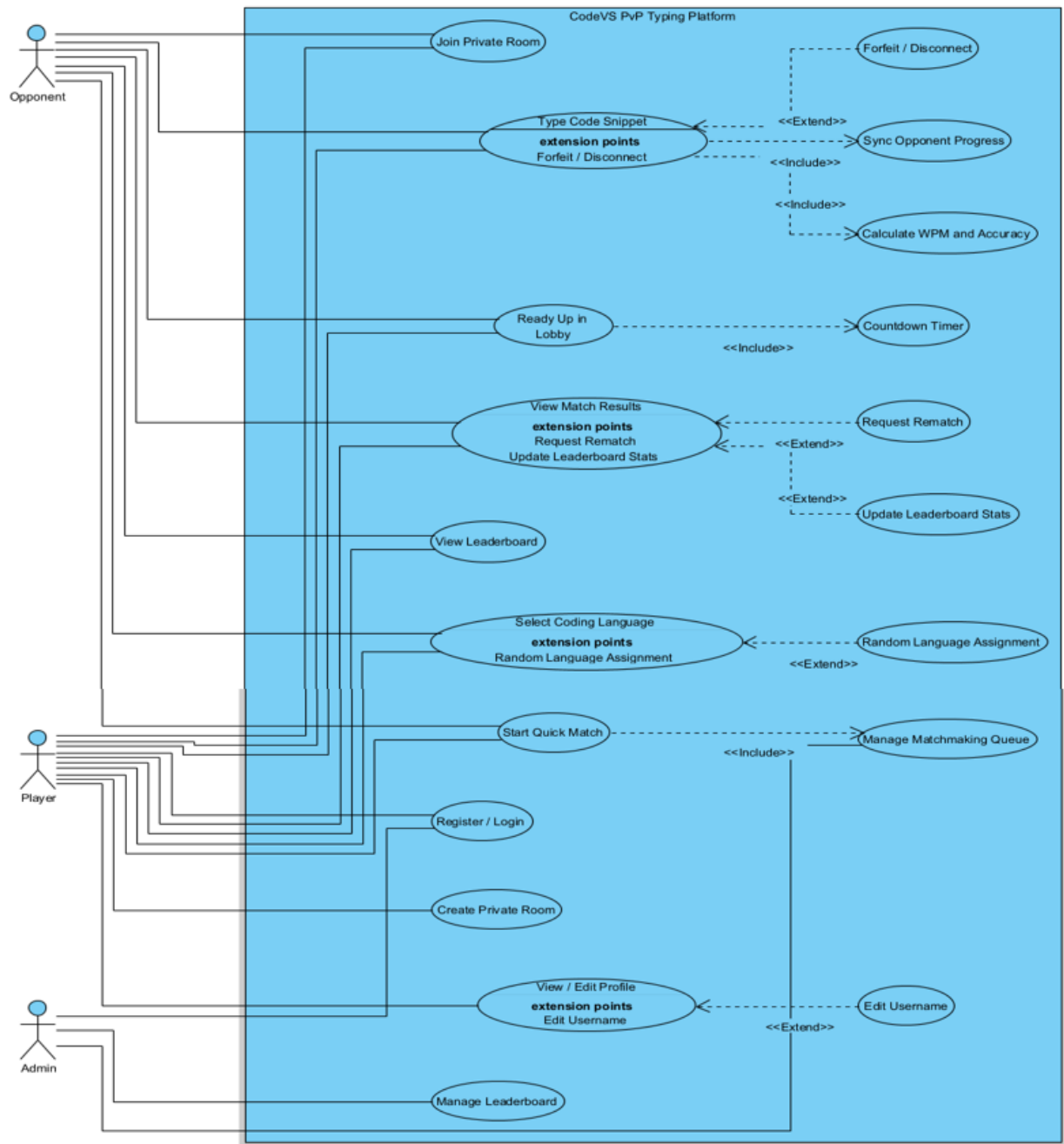
3.3 Sequence Diagram

Shows the interaction timeline between the Client (React), Firebase Authentication, Firestore, and the Realtime Database during a complete multiplayer match session.



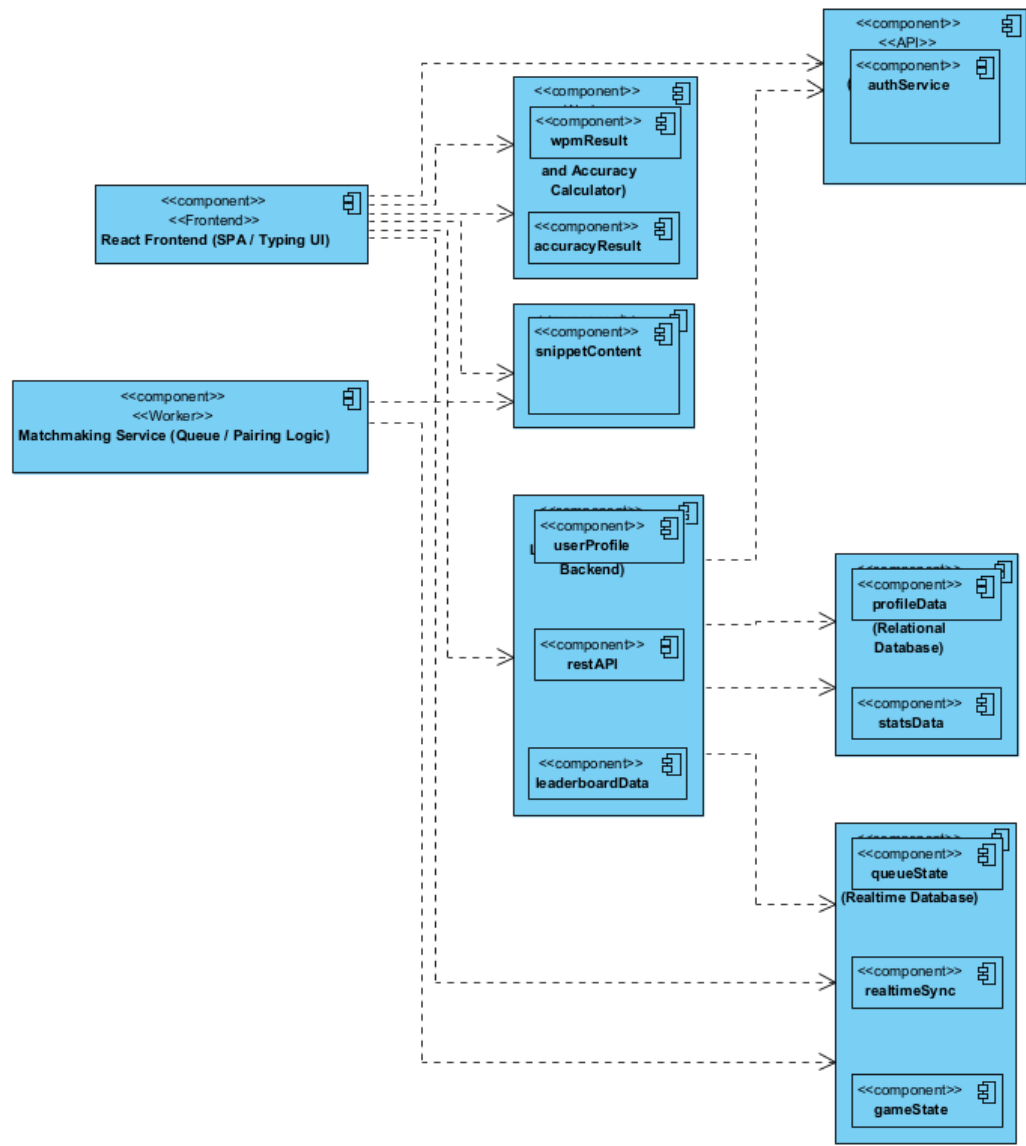
3.4 Use Case Diagram

Identifies all actors (Guest User, Authenticated Player, System/Firebase) and their interactions with the key use cases of the system.



3.5 Component Diagram

Represents the high-level component boundaries: the React SPA, the Firebase Authentication module, Firestore collections, Realtime Database nodes, and the Firebase Hosting infrastructure.

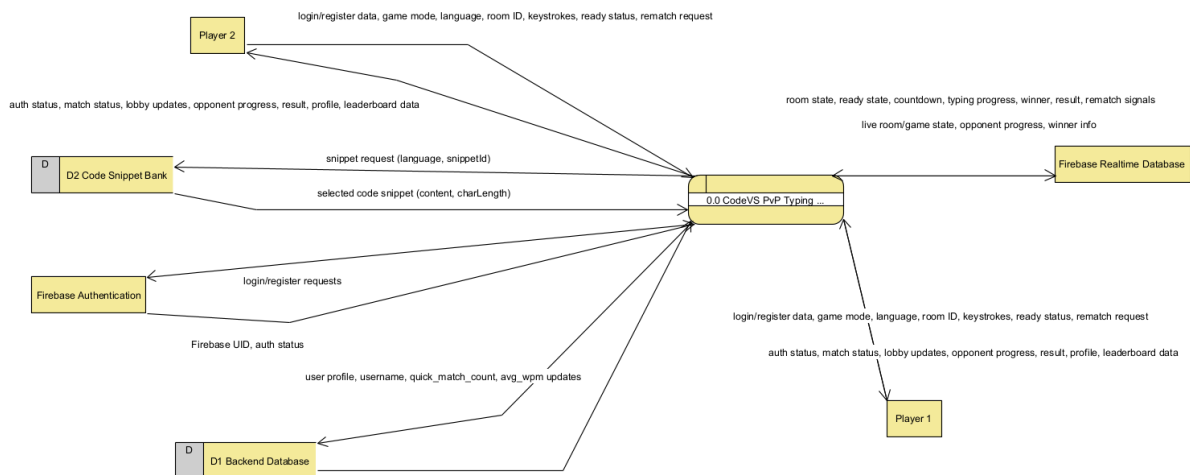


3.6 Data Flow Diagrams (DFD)

The Data Flow Diagrams capture how data moves between the external entities, processes, and data stores in the CodeVS system. Three levels of DFD are provided:

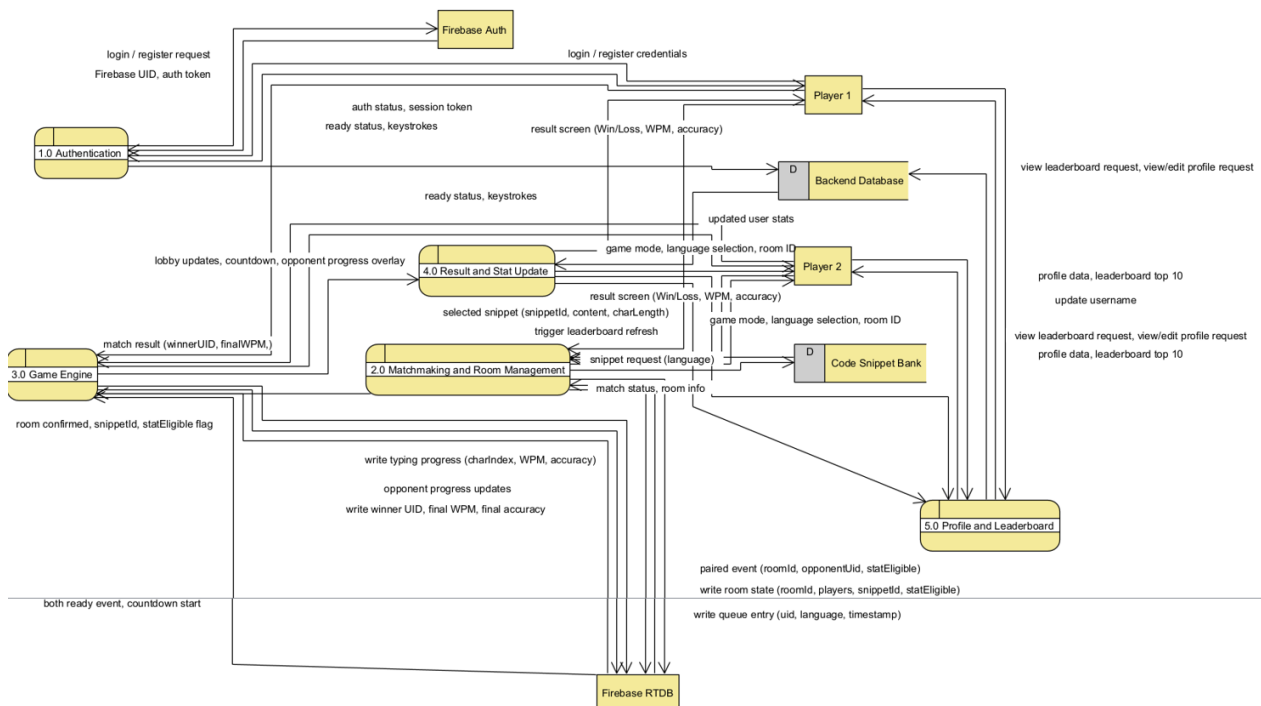
Level 0 — Context Diagram

A single process representing the entire CodeVS system, surrounded by external entities: Player, Firebase Authentication service, and the Firestore/Realtime Database data store.



Level 1 — Main Processes

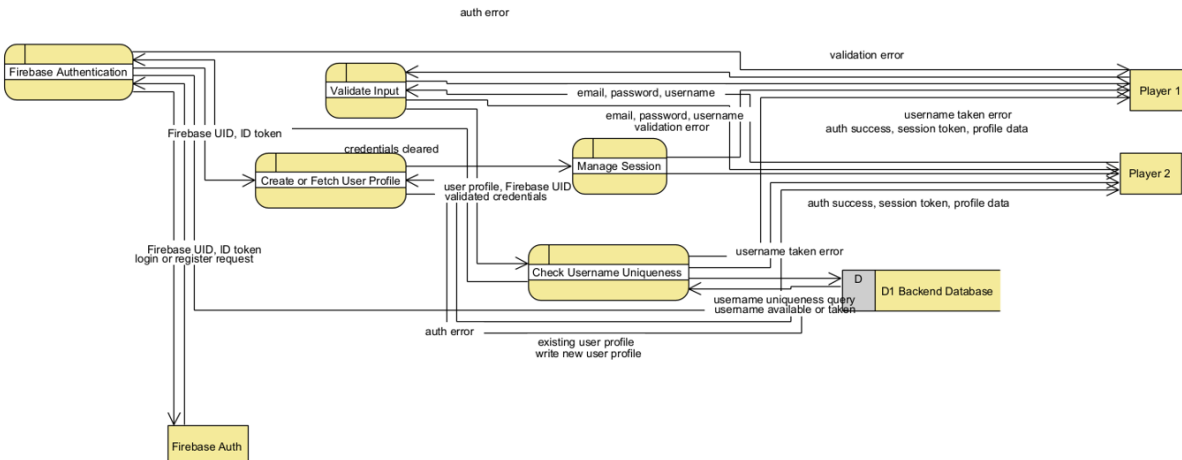
Decomposes the context into five major processes: User Authentication, Matchmaking, Game Session Management, Score Calculation, and Leaderboard Management.



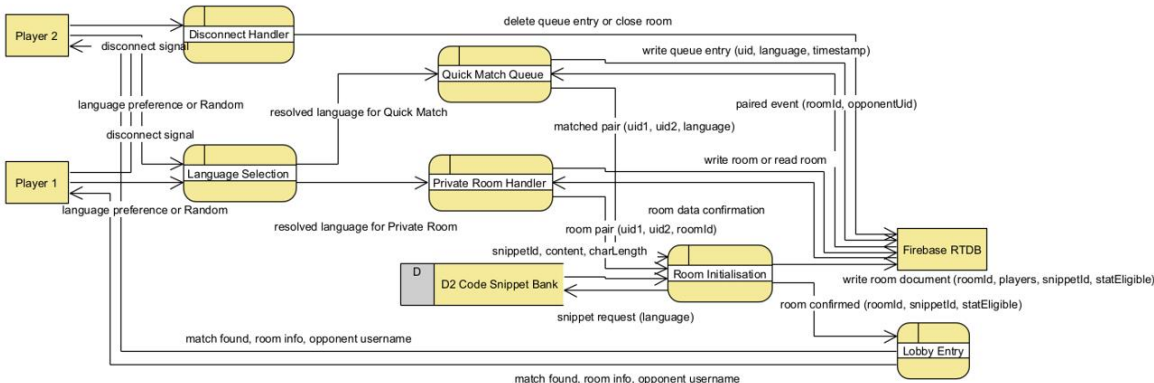
Level 2 — Sub-Process Detail

Further decomposes the Matchmaking and Game Session Management processes, showing internal data flows between room creation, player-ready state, snippet assignment, and typing validation.

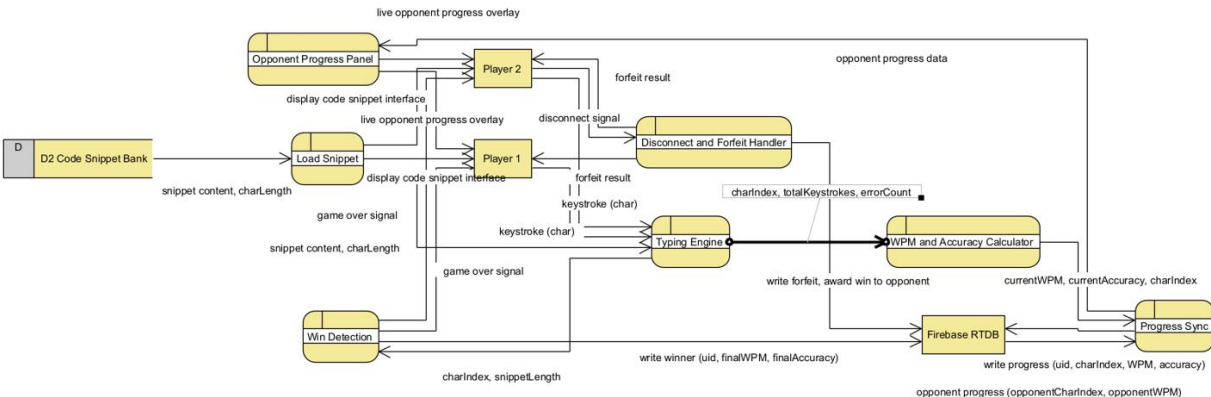
P1 Authentication



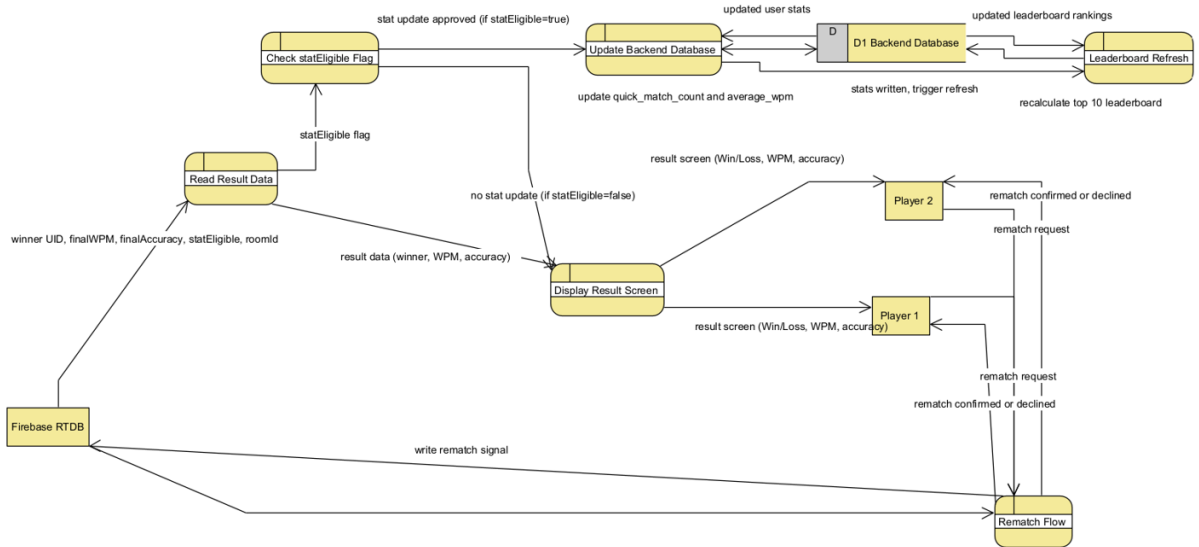
P2: Matchmaking and Room Management



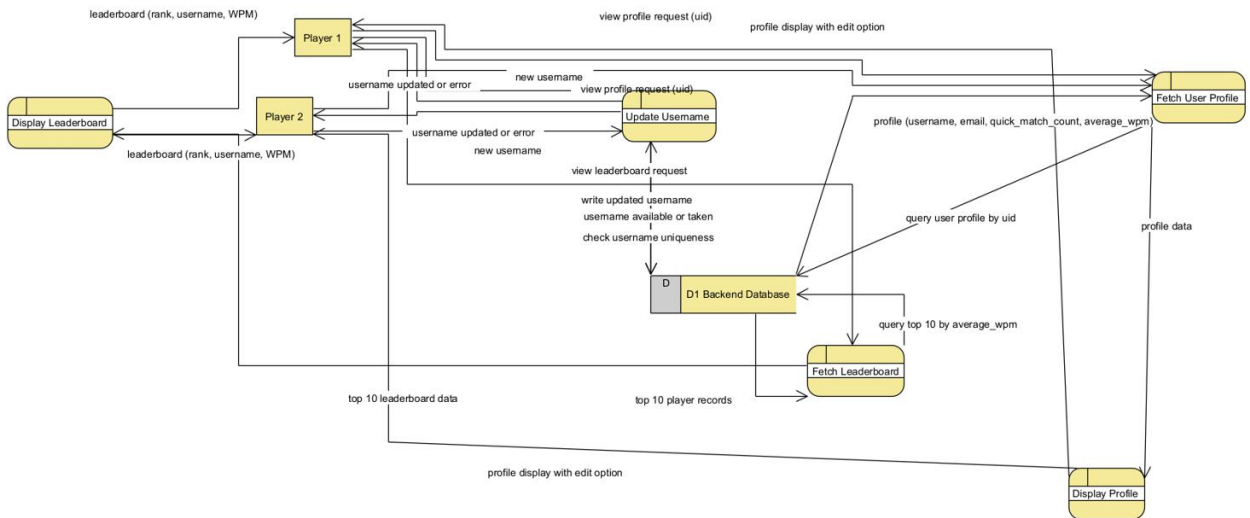
P3: Game Engine



P4: Result and Stat Update



P5: Profile and Leaderboard



4. Project Management Approach

The CodeVS project adopted Agile Scrum as its core software development methodology. All project management activities were coordinated through Jira Cloud (project key: SCRUM), with the GitHub repository linked directly to Jira for automatic branch-to-ticket traceability. Work was organised into Epics, broken down into Tasks, and further decomposed into Subtasks. Three time-boxed Sprints were executed sequentially from project inception to production deployment.

4.1 Team Roles & Responsibilities

Team Member	Role	Responsibilities
Rafsan Riasat	Scrum Master	Project scaffolding, Firebase project initialisation, React router setup, Scrum ceremony facilitation, cross-cutting QA coordination, Firebase security rule review, and final production deployment.
Utsa Roy	Developer — Backend & Game Engine	Firebase Authentication integration, Firestore schema design, Matchmaking queue logic (Quick Match), Room Creation & Join flows, Ready-state synchronisation, WPM & Accuracy engine, Realtime opponent progress tracking, Leaderboard backend, Rematch state machine, Language-picker logic, and edge-case/error handling.
Md Adib Raian	Developer — Frontend & UI	Home screen UI, Global theme & font configuration, Lobby screen UI, Game screen UI, Result screen design, Leaderboard visual overhaul, Mobile responsiveness, and Code Snippet Bank population (C, Python, JS, PHP, C++, Java).

Users

Invite users Approve requests 0 ...

Find people across all your directories and any accounts you manage. [About managed accounts and external users](#)

Total users3

Active users3

Managed accounts ⓘ0

Organization admins1

Q Search by name or email

Account type ▾

Role ▾

Apps ▾

More +

Reset

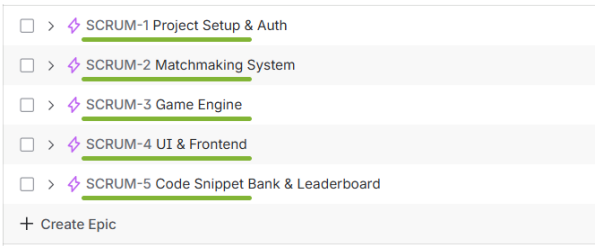
Showing results ↻

User	Status	Last seen ⓘ	Actions
MR Md Adib Raian raian2207020@stud.kuet.ac.bd	ACTIVE	Jun 08, 2026	...
RAFSAN RIASAT (2207006_Rafsan) riasat2207006@stud.kuet.ac.bd - Organization admin	ACTIVE	Jul 13, 2026	...
UR Utsa Roy roy2207027@stud.kuet.ac.bd	ACTIVE	Jul 10, 2026	...

4.2 Epics Overview

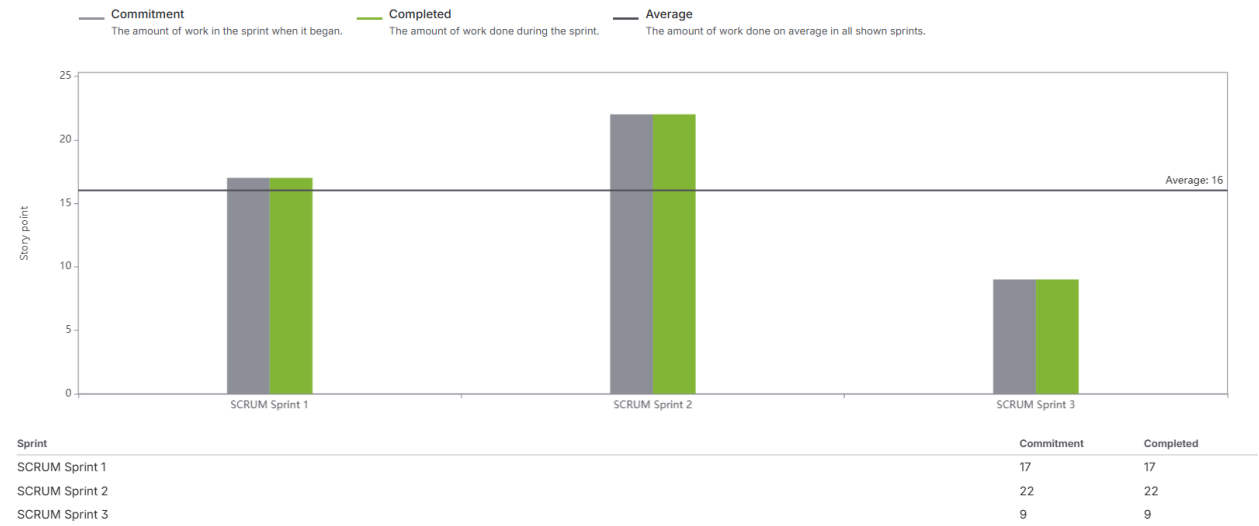
Five high-level Epics were defined to represent the major product domains. Each Epic encapsulates a coherent set of Tasks and Subtasks that were tracked and prioritised across the three sprints.

Epic Key	Epic Name	Scope Summary	Tasks	Status
SCRUM-1	Project Setup & Auth	Initial repository, Firebase project, React scaffold, authentication flows, and Firestore schema.	4	Done
SCRUM-2	Matchmaking System	Quick-match waiting queue, custom room creation (6-char Room ID), and player room-join workflow.	3	Done
SCRUM-3	Game Engine	Ready-state synchronisation, WPM & Accuracy calculation, real-time opponent status, rematch loop, language picker, and error/edge-case hardening.	6	Done
SCRUM-4	UI & Frontend	All screen UIs: Home, Lobby, Game, Result, Leaderboard — including global theming and mobile responsiveness.	8	Done
SCRUM-5	Code Snippet Bank & Leaderboard	Snippet curation and seeding for six languages, leaderboard persistence, ranking queries, and final testing & deployment.	5	Done



5. Sprint Lifecycle

The project was delivered in three sequential Scrum sprints. Each sprint had a defined goal, a curated backlog drawn from the Jira Sprint board, and concluded with a review and retrospective. All 113 issues across the three sprints were resolved with a final status of Done.



5.1 Sprint 1 — Project Foundation

Sprint 1 established the foundational scaffolding for the entire application. The goal was to have a deployable shell with working authentication, a routing structure, an initialised Firebase project, a designed Firestore schema, the home screen layout, global theme, and the first batch of code snippets in the snippet bank.

Issue	Summary	Assignee	Status
SCRUM-6	Firebase project setup	Rafsan Riasat	Done
SCRUM-7	React project setup and Firebase connection	Rafsan Riasat	Done
SCRUM-8	Firebase Authentication Integration	Utsa Roy	Done
SCRUM-9	Design the Firestore database schema	Utsa Roy	Done
SCRUM-10	Quick match — waiting queue logic	Utsa Roy	Done
SCRUM-11	Setup page routing in React	Rafsan Riasat	Done
SCRUM-12	Home screen UI (+3 subtasks)	Md Adib Raian	Done
SCRUM-13	Setup global theme styles and fonts (+4 subtasks)	Md Adib Raian	Done
SCRUM-14	Add C and Python snippets to snippet bank (+4 subtasks)	Md Adib Raian	Done

Total Issues: 20 (9 Tasks + 11 Subtasks)

Date	Event	Work item	Completed	Scope
Sat, Jun 06 2026, 1:31am	Sprint started	SCRUM-8 Firebase Authentication Integration SCRUM-7 React project setup and firebase connection. SCRUM-11 Setup page routing in React SCRUM-9 Design the Firestore database schema SCRUM-10 Quick match — waiting queue logic SCRUM-6 Firebase project setup SCRUM-13 Setup global theme styles and fonts. SCRUM-12 Home screen UI SCRUM-14 Add C and python codes in the code snippet bank.	0	17
Sun, Jun 07 2026, 9:40pm	Work item completed	SCRUM-6 Firebase project setup	0 → 2	17
Sun, Jun 07 2026, 9:40pm	Work item completed	SCRUM-7 React project setup and firebase connection.	2 → 4	17
Sun, Jun 07 2026, 9:40pm	Work item completed	SCRUM-11 Setup page routing in React	4 → 6	17
Sun, Jun 07 2026, 10:22pm	Work item completed	SCRUM-8 Firebase Authentication Integration	6 → 9	17
Sun, Jun 07 2026, 10:26pm	Work item completed	SCRUM-9 Design the Firestore database schema	9 → 12	17
Sun, Jun 07 2026, 11:26pm	Work item completed	SCRUM-10 Quick match — waiting queue logic	12 → 17	17
Sun, Jun 07 2026, 11:33pm	Sprint completed	SCRUM-8 Firebase Authentication Integration SCRUM-7 React project setup and firebase connection. SCRUM-11 Setup page routing in React SCRUM-9 Design the Firestore database schema SCRUM-10 Quick match — waiting queue logic SCRUM-6 Firebase project setup SCRUM-13 Setup global theme styles and fonts. SCRUM-12 Home screen UI SCRUM-14 Add C and python codes in the code snippet bank.	17	17

5.2 Sprint 2 — Core Multiplayer Engine

Sprint 2 delivered the working multiplayer core. By the end of this sprint, two authenticated players could create or join a private room, reach a lobby, ready up, see a countdown, and engage in a live simultaneous typing race with real-time opponent progress visible to both sides. The leaderboard persistence layer was also completed.

Issue	Summary	Assignee	Status
SCRUM-15	Room Creation Logic (+5 subtasks)	Utsa Roy	Done
SCRUM-16	Room Join Logic (+5 subtasks)	Utsa Roy	Done
SCRUM-17	Ready State Logic (+5 subtasks)	Utsa Roy	Done
SCRUM-18	WPM & Accuracy Calculation	Utsa Roy	Done
SCRUM-19	Leaderboard Logic (+5 subtasks)	Utsa Roy	Done
SCRUM-20	Lobby Screen UI (+3 subtasks)	Md Adib Raian	Done
SCRUM-21	Game Screen UI (+5 subtasks)	Md Adib Raian	Done
SCRUM-22	Real-time Opponent Status (+4 subtasks)	Utsa Roy	Done
SCRUM-23	Add JavaScript & PHP snippets (+4 subtasks)	Md Adib Raian	Done

Total Issues: 45 (9 Tasks + 36 Subtasks)

Date	Event	Work item	Completed	Scope
Sun, Jun 07 2026, 11:34pm	Sprint started	SCRUM-15 Room Creation Logic SCRUM-17 Ready State Logic SCRUM-16 Room Join Logic SCRUM-20 Lobby Screen UI SCRUM-22 Real-time Opponent Status SCRUM-21 Game Screen UI SCRUM-23 Add JavaScript and PHP codes in the Code Snippet Bank SCRUM-19 Leaderboard Logic SCRUM-18 WPM & Accuracy Calculation	0	22
Sun, Jun 07 2026, 11:51pm	Work item completed	SCRUM-15 Room Creation Logic	0 → 3	22
Sun, Jun 07 2026, 11:55pm	Work item completed	SCRUM-16 Room Join Logic	3 → 8	22
Mon, Jun 08 2026, 12:06am	Work item completed	SCRUM-17 Ready State Logic	8 → 11	22
Mon, Jun 08 2026, 12:06am	Work item completed	SCRUM-22 Real-time Opponent Status	11 → 14	22
Mon, Jun 08 2026, 12:06am	Work item completed	SCRUM-18 WPM & Accuracy Calculation	14 → 19	22
Mon, Jun 08 2026, 12:43am	Work item completed	SCRUM-19 Leaderboard Logic	19 → 22	22
Mon, Jun 08 2026, 12:47am	Work item re-opened	SCRUM-15 Room Creation Logic	22 → 19	22
Mon, Jun 08 2026, 12:47am	Work item completed	SCRUM-15 Room Creation Logic SCRUM-17 Ready State Logic SCRUM-16 Room Join Logic SCRUM-20 Lobby Screen UI SCRUM-22 Real-time Opponent Status SCRUM-21 Game Screen UI SCRUM-23 Add JavaScript and PHP codes in the Code Snippet Bank SCRUM-19 Leaderboard Logic SCRUM-18 WPM & Accuracy Calculation	19 → 22	22
Mon, Jun 08 2026, 12:48am	Sprint completed		22	22

5.3 Sprint 3 — Polish, Testing & Deployment

Sprint 3 focused on product completeness. The rematch feature, language-picker logic, additional snippet languages (C++ and Java), visual overhauling of the result and leaderboard screens, mobile responsiveness, comprehensive error handling for disconnects and race conditions, and full QA + production deployment were all completed in this sprint.

Issue	Summary	Assignee	Status
SCRUM-24	Result Screen Design (+4 subtasks)	Md Adib Raian	Done
SCRUM-25	Visual Overhaul of Leaderboard Screen (+4 subtasks)	Md Adib Raian	Done
SCRUM-26	Rematch Button Logic (+5 subtasks)	Utsa Roy	Done
SCRUM-27	Language Picker Logic (+5 subtasks)	Utsa Roy	Done
SCRUM-28	Error Handling & Edge Cases (+5 subtasks)	Utsa Roy	Done
SCRUM-29	Mobile Responsiveness (+4 subtasks)	Md Adib Raian	Done
SCRUM-30	Add C++ & Java codes to Snippet Bank (+4 subtasks)	Md Adib Raian	Done
SCRUM-31	Final Testing & Deployment (+4 subtasks)	Rafsan Riasat	Done

Total Issues: 43 (8 Tasks + 35 Subtasks)

Date	Event	Work item	Completed	Scope
Mon, Jun 08 2026, 12:49am	Sprint started	SCRUM-30 Add C++ & Java codes to the Code Snippet Bank SCRUM-24 Result Screen Design SCRUM-26 Rematch Button Logic SCRUM-25 Visual Overhaul of Leaderboard Screen SCRUM-28 Error Handling & Edge Cases SCRUM-27 Language Picker Logic SCRUM-29 Mobile Responsiveness SCRUM-31 Final Testing & Deployment	0	9
Mon, Jun 08 2026, 1:11am	Work item completed	SCRUM-26 Rematch Button Logic	0 → 2	9
Mon, Jun 08 2026, 1:11am	Work item completed	SCRUM-27 Language Picker Logic	2 → 4	9
Mon, Jun 08 2026, 1:27am	Work item completed	SCRUM-28 Error Handling & Edge Cases	4 → 7	9
Mon, Jun 08 2026, 2:37am	Work item completed	SCRUM-31 Final Testing & Deployment	7 → 9	9
Mon, Jun 08 2026, 2:42am	Work item re-opened	SCRUM-31 Final Testing & Deployment	9 → 7	9
Mon, Jul 13 2026, 6:48pm	Work item completed	SCRUM-31 Final Testing & Deployment	7 → 9	9
Mon, Jul 13 2026, 6:49pm	Sprint completed	SCRUM-30 Add C++ & Java codes to the Code Snippet Bank SCRUM-24 Result Screen Design SCRUM-26 Rematch Button Logic SCRUM-25 Visual Overhaul of Leaderboard Screen SCRUM-28 Error Handling & Edge Cases SCRUM-27 Language Picker Logic SCRUM-29 Mobile Responsiveness SCRUM-31 Final Testing & Deployment	9	9

6. Detailed Task Breakdown by Epic

This section enumerates every Task and its Subtasks, grouped by Epic, providing full traceability from the high-level product backlog down to the individual atomic work items tracked in Jira.

6.1 Epic SCRUM-1 — Project Setup & Auth

SCRUM-6 — Firebase Project Setup

Assignee: Rafsan Riasat

Status: Done

Sprint: Sprint 1

Created the Firebase project on the Firebase console, configured API keys, enabled Firestore and Realtime Database, and set initial security rules.

SCRUM-7 — React Project Setup & Firebase Connection

Assignee: Rafsan Riasat

Status: Done

Sprint: Sprint 1

Bootstrapped the React application using Vite, installed all required dependencies (React Router, Firebase SDK), and configured the firebase.js initialisation module.

SCRUM-8 — Firebase Authentication Integration

Assignee: Utsa Roy

Status: Done

Sprint: Sprint 1

Implemented full Login and Registration flows using Firebase Auth (email/password). This task was developed in the dedicated branch `SCRUM-8-firebase-authentication-integration` and merged via Pull Request #9.

SCRUM-9 — Design the Firestore Database Schema

Assignee: Utsa Roy

Status: Done

Sprint: Sprint 1

Designed and documented the Firestore collection structure: users, rooms, snippets, snippet_metadata, and leaderboard collections, establishing data contracts for all subsequent backend tasks.

6.2 Epic SCRUM-2 — Matchmaking System

SCRUM-10 — Quick Match — Waiting Queue Logic

Assignee: Utsa Roy

Status: Done

Sprint: Sprint 1

Implemented the global matchmaking queue using the Firebase Realtime Database. Players who click 'Quick Match' are placed in a presence-tracked queue; the system pairs the first two waiting players and creates a room automatically.

SCRUM-15 — Room Creation Logic

Assignee: Utsa Roy

Status: Done

Sprint: Sprint 2

Implemented the full custom room creation flow allowing a host player to generate a unique 6-character alphanumeric Room ID, initialise the room in Firestore, and wait in the lobby.

Subtasks:

Subtask Key	Summary	Assignee	Status
SCRUM-32	Generate unique 6-character room IDs	Utsa Roy	Done
SCRUM-33	Firebase room schema and data model	Utsa Roy	Done
SCRUM-34	Host player initialisation in room creation	Utsa Roy	Done
SCRUM-35	Room validation and error handling for room creation	Utsa Roy	Done
SCRUM-36	Room creation unit testing and verification	Rafsan Riasat	Done

SCRUM-16 — Room Join Logic

Assignee: Utsa Roy

Status: Done

Sprint: Sprint 2

Implemented the join flow: the guest player enters a 6-character Room ID, the system validates the room's existence and availability, then adds the player to the room document and triggers lobby synchronisation for both players. Branch: `SCRUM-16-room-join-logic`.

Subtasks:

Subtask Key	Summary	Assignee	Status
SCRUM-37	Room ID validation for joining flow	Utsa Roy	Done
SCRUM-38	Player join workflow and room state update	Utsa Roy	Done
SCRUM-39	Real-time presence tracking for joined players	Utsa Roy	Done
SCRUM-40	Lobby synchronisation after player joins	Utsa Roy	Done
SCRUM-41	Join flow end-to-end testing and validation	Rafsan Riasat	Done

6.3 Epic SCRUM-3 — Game Engine

SCRUM-17 — Ready State Logic

Assignee: Utsa Roy

Status: Done

Sprint: Sprint 2

Players in the lobby each click 'Ready'. The system tracks both ready states in the Realtime Database. When both are ready, a 3-second countdown is displayed and the game transitions to the active typing phase. Disconnects during the lobby phase are handled gracefully.

Subtasks:

Subtask Key	Summary	Assignee	Status
SCRUM-42	Player ready state synchronisation via realtime database	Utsa Roy	Done
SCRUM-43	Lobby ready status UI indicators and state display	Rafsan Riasat	Done
SCRUM-44	Countdown timer before game start when all players are ready	Utsa Roy	Done
SCRUM-45	Trigger game start when countdown completes	Utsa Roy	Done
SCRUM-46	Handle player disconnects and unready state during lobby phase	Rafsan Riasat	Done

SCRUM-18 — WPM & Accuracy Calculation

Assignee: Utsa Roy

Status: Done

Sprint: Sprint 2

Implemented the scoring engine. WPM is calculated as (correct characters / 5) / elapsed minutes. Accuracy is (correct keystrokes / total keystrokes) × 100%. Both metrics are computed client-side and persisted to Firestore upon game completion.

SCRUM-22 — Real-time Opponent Status

Assignee: Utsa Roy

Status: Done

Sprint: Sprint 2

The typing progress index (current character position) for each player is written to the Realtime Database on every keystroke. A Firebase listener on the opponent's node updates the opponent progress bar in real time, with write optimisation to reduce Firebase costs.

Subtasks:

Subtask Key	Summary	Assignee	Status
SCRUM-52	Define and sync opponent typing progress data model	Utsa Roy	Done
SCRUM-53	Implement Firebase listener for real-time opponent progress updates	Utsa Roy	Done
SCRUM-54	Optimise update frequency for opponent progress (reduce Firebase writes)	Utsa Roy	Done
SCRUM-55	Handle opponent disconnect/reconnect state during gameplay	Utsa Roy	Done

SCRUM-26 — Rematch Button Logic

Assignee: Utsa Roy

Status: Done

Sprint: Sprint 3

After match completion, either player can request a rematch. Both players must accept; the system creates a new game session and resets all match state. If either player declines, both are navigated to the home screen. Branch: `SCRUM-26-rematch-button-logic`.

Subtasks:

Subtask Key	Summary	Assignee	Status
SCRUM-56	Rematch request button UI and interaction flow	Utsa Roy	Done
SCRUM-57	Rematch request state management and synchronisation between players	Utsa Roy	Done
SCRUM-58	Accept or decline rematch request flow and UI handling	Utsa Roy	Done
SCRUM-59	Create new game session and reset match state for rematch	Utsa Roy	Done
SCRUM-60	Test rematch flow including edge cases and reconnections	Rafsan Riasat	Done

SCRUM-27 — Language Picker Logic

Assignee: Utsa Roy

Status: Done

Sprint: Sprint 3

Players select a programming language on the home screen before matchmaking. The preference is persisted and used to route the player into a language-specific matchmaking queue. Snippets are drawn from the appropriate language pool in Firestore.

Subtasks:

Subtask Key	Summary	Assignee	Status
SCRUM-61	Build language selection UI for player preferences	Md Adib Raian	Done
SCRUM-62	Store and persist user language preference for matchmaking	Utsa Roy	Done
SCRUM-63	Implement language-based matchmaking logic between players	Utsa Roy	Done
SCRUM-64	Fetch and use language-specific snippet pools for gameplay	Utsa Roy	Done
SCRUM-65	Test language system including matchmaking and snippet distribution	Rafsan Riasat	Done

SCRUM-28 — Error Handling & Edge Cases

Assignee: Utsa Roy

Status: Done

Sprint: Sprint 3

A comprehensive stabilisation pass was performed to handle room flow edge cases (disconnects, invalid states, race conditions), multiplayer desync recovery, user-facing error messages, and cross-browser compatibility. Branch: `SCRUM-28-error-handling-edge-cases`.

Subtasks:

Subtask Key	Summary	Assignee	Status
SCRUM-66	Handle room flow edge cases (disconnects, invalid states, race conditions)	Utsa Roy	Done
SCRUM-67	Handle multiplayer synchronisation failure recovery (desync,	Utsa Roy	Done

	latency, retry logic)		
SCRUM-68	Implement user-facing error messages and recovery UI for failed game states	Utsa Roy	Done
SCRUM-69	Cross-browser and device compatibility testing for full game flow	Utsa Roy	Done
SCRUM-70	Bug-fix sprint pass and stabilisation for production readiness	Rafsan Riasat	Done

6.4 Epic SCRUM-4 — UI & Frontend

SCRUM-11 — Setup Page Routing in React

Assignee: Rafsan Riasat

Status: Done

Sprint: Sprint 1

Configured React Router v6 with protected routes (requiring authentication) for the Home, Lobby, Game, Result, and Leaderboard pages, and public routes for Login/Register.

SCRUM-12 — Home Screen UI

Assignee: Md Adib Raian

Status: Done

Sprint: Sprint 1

Implemented the landing screen with a hacker-themed dark aesthetic, animated background, Quick Match, Create Room, and Join Room action buttons, and the language selector.

Subtasks:

Subtask Key	Summary	Assignee	Status
SCRUM-88	Design quick match, create room, and join room buttons	Md Adib Raian	Done
SCRUM-89	Integrate cool animations into home screen UI	Md Adib Raian	Done
SCRUM-90	Implement hacker-themed UI layout for home screen	Md Adib Raian	Done

SCRUM-13 — Setup Global Theme Styles & Fonts

Assignee: Md Adib Raian

Status: Done

Sprint: Sprint 1

Established the global design system: a dark hacker/cyber aesthetic, font selection (monospaced coding font), global CSS variables, and cross-browser verification.

Subtasks:

Subtask Key	Summary	Assignee	Status
SCRUM-91	Establish global style guidelines	Md Adib Raian	Done
SCRUM-92	Configure theme provider or global CSS	Md Adib Raian	Done
SCRUM-93	Test theme styles across major browsers and devices	Md Adib Raian	Done
SCRUM-94	Select and implement preferred font	Md Adib Raian	Done

SCRUM-20 — Lobby Screen UI

Assignee: Md Adib Raian

Status: Done

Sprint: Sprint 2

Implemented the lobby screen with ready-button UI for both players, a countdown animation overlay, and responsive dark-theme layout. Branch: `SCRUM-20-lobby-screen-ui`.

Subtasks:

Subtask Key	Summary	Assignee	Status
SCRUM-95	Implement ready button functionality for both players	Md Adib Raian	Done
SCRUM-96	Add countdown timer UI and animation for game start	Md Adib Raian	Done
SCRUM-97	Ensure responsive layout and dark theme styling for lobby screen	Md Adib Raian	Done

SCRUM-21 — Game Screen UI

Assignee: Md Adib Raian

Status: Done

Sprint: Sprint 2

Designed the primary game interface: a split-panel layout with the active player's typing area on one side and the opponent's read-only overlay and progress bar on the other, styled with the hacker/cyber dark theme. Branch: `SCRUM-21-game-screen-ui`.

Subtasks:

Subtask Key	Summary	Assignee	Status
SCRUM-98	Implement side-by-side layout for player and opponent panels	Md Adib Raian	Done
SCRUM-99	Integrate typing input area for active player	Md Adib Raian	Done
SCRUM-100	Display opponent progress as read-only overlay	Md Adib Raian	Done
SCRUM-101	Apply hacker/cyber dark theme styling to game screen	Md Adib Raian	Done
SCRUM-102	Ensure responsive design for game screen UI	Md Adib Raian	Done

SCRUM-24 — Result Screen Design

Assignee: Md Adib Raian

Status: Done

Sprint: Sprint 3

Built the animated post-match result screen revealing the winner, WPM/Accuracy for both players, and action buttons for Rematch and Return to Home. Branch: `SCRUM-24-result-screen-design`.

Subtasks:

Subtask Key	Summary	Assignee	Status
SCRUM-103	Implement result screen UI in React	Md Adib Raian	Done
SCRUM-104	Apply hacker/cyber dark theme styling to result screen	Md Adib Raian	Done
SCRUM-105	Ensure responsive design for result screen UI	Md Adib Raian	Done
SCRUM-106	Add animations for result reveal and transitions	Md Adib Raian	Done

SCRUM-25 — Visual Overhaul of Leaderboard Screen

Assignee: Md Adib Raian

Status: Done

Sprint: Sprint 3

Overhauled the leaderboard page with a responsive layout, consistent dark-theme styling, ranking animations, and preservation of existing leaderboard data integrity. Branch: `SCRUM-25-leaderboard-visual-overhaul`.

Subtasks:

Subtask Key	Summary	Assignee	Status
SCRUM-111	Implement responsive layout for leaderboard screen	Md Adib Raian	Done
SCRUM-112	Apply dark theme styling to leaderboard screen	Md Adib Raian	Done
SCRUM-113	Add animations and transitions to leaderboard elements	Md Adib Raian	Done
SCRUM-114	Verify preservation of existing leaderboard functionality	Md Adib Raian	Done

SCRUM-29 — Mobile Responsiveness

Assignee: Md Adib Raian

Status: Done

Sprint: Sprint 3

Comprehensive mobile-first pass across all screens. Minimalistic layout for small viewports, touch-optimised button sizes, and navigation simplification for mobile users. Branch: `SCRUM-29-mobile-responsiveness`.

Subtasks:

Subtask Key	Summary	Assignee	Status
SCRUM-107	Design minimalistic game screen layout for mobile	Md Adib Raian	Done
SCRUM-108	Optimise touch controls and button sizes for mobile	Md Adib Raian	Done
SCRUM-109	Test and refine mobile UI responsiveness across devices	Md Adib Raian	Done
SCRUM-110	Streamline navigation and minimise clutter for mobile users	Md Adib Raian	Done

6.5 Epic SCRUM-5 — Code Snippet Bank & Leaderboard

SCRUM-14 — Add C and Python Snippets to the Snippet Bank

Assignee: Md Adib Raian

Status: Done

Sprint: Sprint 1

Curated and formatted ten code snippets each for the C and Python languages and seeded them into the Firestore `snippets` and `snippet\_metadata` collections. Verified visibility in the UI.

Subtasks:

Subtask Key	Summary	Assignee	Status
SCRUM-75	Curate and format ten Python code snippets for the snippet bank	Md Adib Raian	Done
SCRUM-76	Integrate C and Python snippets into the code snippet bank data store	Md Adib Raian	Done
SCRUM-77	Update website UI to display snippets during language selection	Md Adib Raian	Done
SCRUM-78	Verify snippet visibility on website and during matchmaking	Md Adib Raian	Done

SCRUM-19 — Leaderboard Logic

Assignee: Utsa Roy

Status: Done

Sprint: Sprint 2

Designed the leaderboard Firestore data model and implemented: post-match result persistence, WPM/accuracy metric computation, and a Firestore query returning the top-ranked players. Branch: `SCRUM-19-leaderboard-logic`.

Subtasks:

Subtask Key	Summary	Assignee	Status
SCRUM-47	Design leaderboard data model and Firestore schema	Utsa Roy	Done
SCRUM-48	Persist match results after each game in Firestore	Utsa Roy	Done
SCRUM-49	Compute WPM, accuracy and performance metrics after match	Utsa Roy	Done
SCRUM-50	Implement top ranking query for leaderboard (top players retrieval)	Utsa Roy	Done
SCRUM-51	Integrate leaderboard UI with backend and add basic testing	Utsa Roy	Done

SCRUM-23 — Add JavaScript & PHP Snippets to the Snippet Bank

Assignee: Md Adib Raian

Status: Done

Sprint: Sprint 2

Extended the snippet bank with JavaScript and PHP code samples, updated the UI language selector, and validated backward compatibility with existing snippets. Branch: `SCRUM-23-add-js-php-snippets`.

Subtasks:

Subtask Key	Summary	Assignee	Status
SCRUM-80	Integrate JavaScript and PHP snippets into the code snippet bank data store	Md Adib Raian	Done
SCRUM-81	Verify snippet visibility and preview for JavaScript and PHP on the website	Md Adib Raian	Done
SCRUM-82	Update website UI to display JavaScript and PHP snippets during language selection	Md Adib Raian	Done
SCRUM-83	Validate existing snippet functionality after new additions	Md Adib Raian	Done

SCRUM-30 — Add C++ & Java Snippets to the Snippet Bank

Assignee: Md Adib Raian

Status: Done

Sprint: Sprint 3

Added C++ and Java snippet pools, completing six-language support. Validated full snippet library integrity. Branch: `SCRUM-30-add-cpp-java-snippets`.

Subtasks:

Subtask Key	Summary	Assignee	Status
SCRUM-84	Integrate C++ and Java snippets into the code snippet bank data store	Md Adib Raian	Done
SCRUM-85	Update website UI to display C++ and Java snippets during language selection	Md Adib Raian	Done

SCRUM-86	Verify snippet visibility and preview for C++ and Java on the website	Md Adib Raian	Done
SCRUM-87	Validate existing snippet functionality after new additions	Md Adib Raian	Done

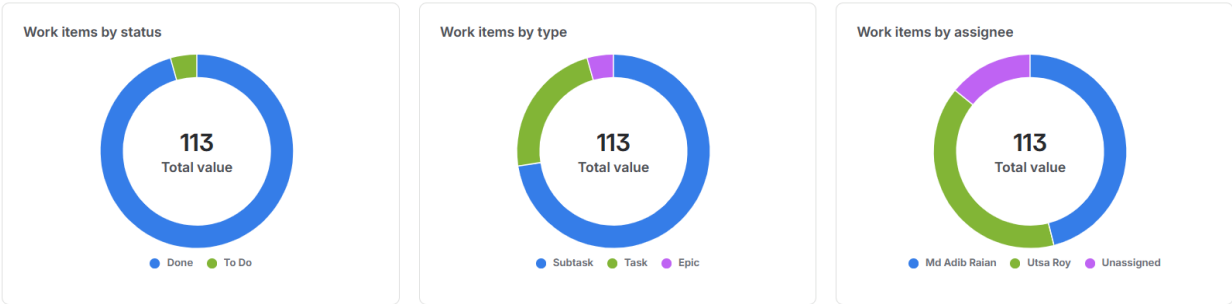
SCRUM-31 — Final Testing & Deployment

Assignee: Rafsan Riasat
Status: Done
Sprint: Sprint 3

Coordinated the final QA pass: created a functional testing checklist covering all game flows, conducted a full multiplayer testing session, reviewed and hardened Firebase security rules, and deployed the production build to Firebase Hosting.

Subtasks:

Subtask Key	Summary	Assignee	Status
SCRUM-71	Create functional testing checklist for full game flow	Rafsan Riasat	Done
SCRUM-72	Conduct full multiplayer testing session with multiple users	Rafsan Riasat	Done
SCRUM-73	Review and harden Firebase security rules before production deployment	Rafsan Riasat	Done
SCRUM-74	Deploy application to production environment (Firebase Hosting setup)	Rafsan Riasat	Done



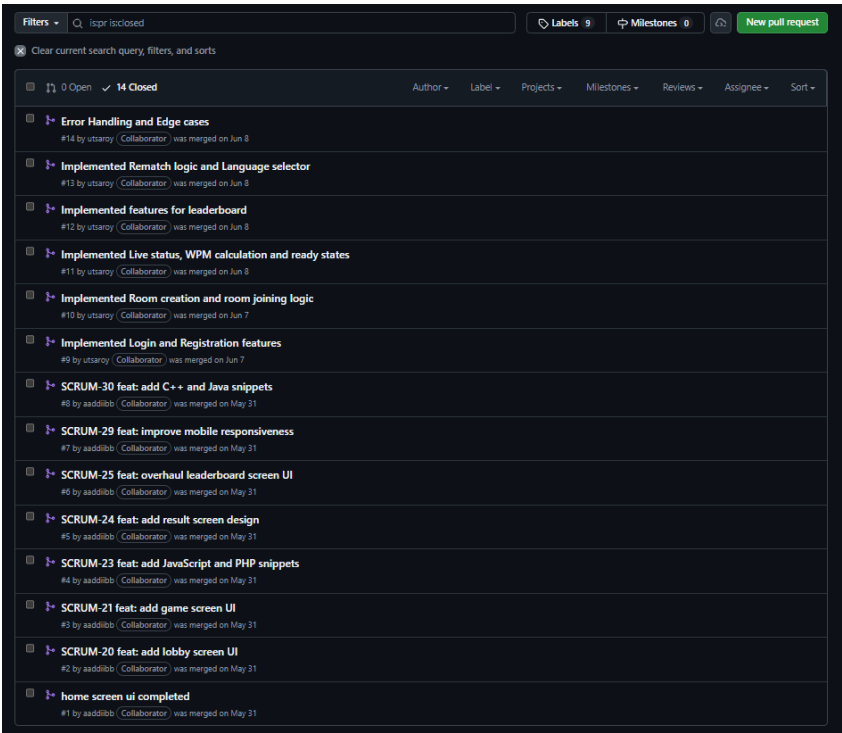
7. Version Control & Git Workflow

The project used Git for version control hosted on GitHub under the repository `narukami00/CodeVS`. The repository was integrated with Jira Cloud so that branches created with the naming convention `SCRUM-{n}-{task-slug}` were automatically linked to the corresponding Jira issue, providing end-to-end traceability from backlog item to code change.

7.1 Branching Strategy

All feature development was performed in dedicated Jira-linked branches. When work was complete, a Pull Request was raised to merge back into `main`. This ensured code review checkpoints and a clean linear history on the main branch. The table below maps every surviving feature branch to its Jira issue.

Branch Name	Jira Ticket	Task Summary	Developer
SCRUM-8-firebase-authentication-integration	SCRUM-8	Firebase Authentication Integration	Utsa Roy
SCRUM-16-room-join-logic	SCRUM-16	Room Join Logic	Utsa Roy
SCRUM-19-leaderboard-logic	SCRUM-19	Leaderboard Logic	Utsa Roy
SCRUM-20-lobby-screen-ui	SCRUM-20	Lobby Screen UI	Md Adib Raian
SCRUM-21-game-screen-ui	SCRUM-21	Game Screen UI	Md Adib Raian
SCRUM-22-real-time-opponent-status	SCRUM-22	Real-time Opponent Status	Utsa Roy
SCRUM-23-add-js-php-snippets	SCRUM-23	Add JS & PHP Snippets	Md Adib Raian
SCRUM-24-result-screen-design	SCRUM-24	Result Screen Design	Md Adib Raian
SCRUM-25-leaderboard-visual-overhaul	SCRUM-25	Visual Overhaul of Leaderboard Screen	Md Adib Raian
SCRUM-26-rematch-button-logic	SCRUM-26	Rematch Button Logic	Utsa Roy
SCRUM-28-error-handling-edge-cases	SCRUM-28	Error Handling & Edge Cases	Utsa Roy
SCRUM-29-mobile-responsiveness	SCRUM-29	Mobile Responsiveness	Md Adib Raian
SCRUM-30-add-cpp-java-snippets	SCRUM-30	Add C++ & Java Snippets	Md Adib Raian
frontend	N/A	Early frontend development (pre-sprint)	Md Adib Raian
main	N/A	Production branch — all merges land here	All



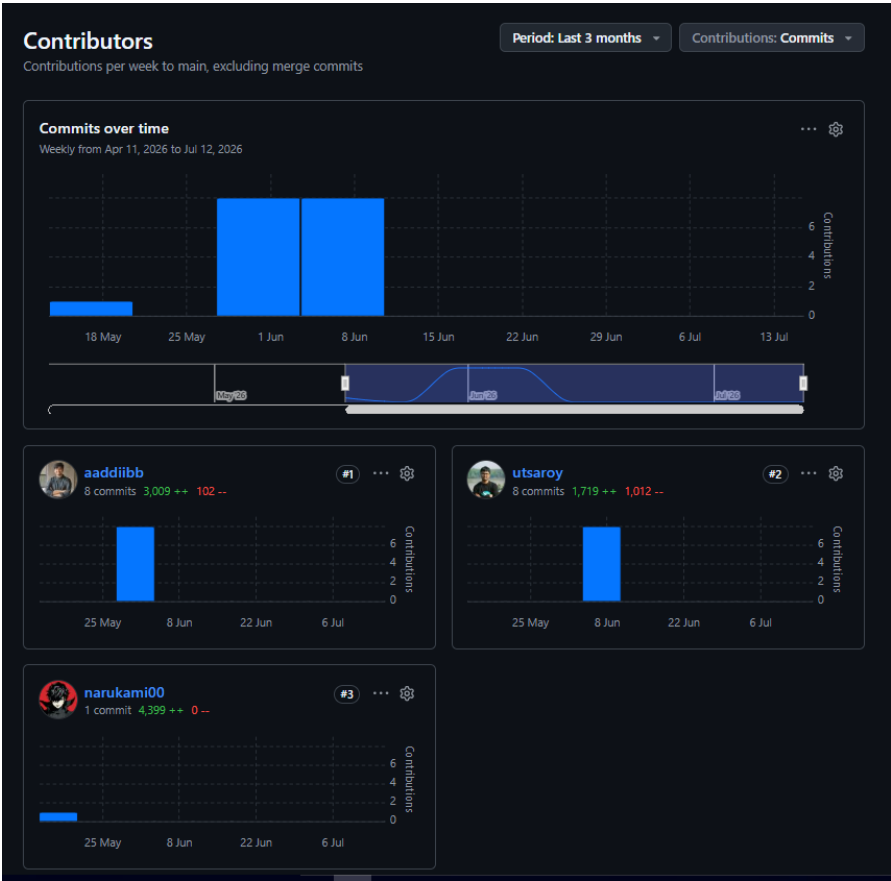
7.2 Commit History Summary

The table below lists all 31 commits in chronological order (oldest to newest) across all branches. 21 out of 31 commits (68%) carry a Jira ticket reference in the commit message, demonstrating strong branch-to-issue traceability. The remaining commits are merge commits or untagged hotfixes.

Date	Author	Commit Message	Short Hash
2026-05-22	Rafsan Riasat	feat: add initial project structure with React, routing, and Firebase integration	f623655b
2026-05-31	aaddiibb	home screen ui completed	f9fe6562
2026-05-31	Adib Raian	Merge PR #1 from narukami00/frontend	a43f488e
2026-05-31	aaddiibb	SCRUM-20 feat: add lobby screen UI	b3374309
2026-05-31	Adib Raian	Merge PR #2 from narukami00/SCRUM-20-lobby-screen-ui	9cd0da59
2026-05-31	aaddiibb	SCRUM-21 feat: add game screen UI	d6e90e8b
2026-05-31	Adib Raian	Merge PR #3 from narukami00/SCRUM-21-game-screen-ui	4fb57c16
2026-05-31	aaddiibb	SCRUM-23 feat: add JavaScript and PHP snippets	403e7d02
2026-05-31	Adib Raian	Merge PR #4 from narukami00/SCRUM-23-add-js-php-snippets	93e1ca02
2026-05-31	aaddiibb	SCRUM-24 feat: add result screen design	f2a9bdcb
2026-05-31	Adib Raian	Merge PR #5 from narukami00/SCRUM-24-result-screen-design	d20f6196
2026-05-31	aaddiibb	SCRUM-25 feat: overhaul leaderboard screen UI	7046b160
2026-05-31	Adib Raian	Merge PR #6 from narukami00/SCRUM-25-leaderboard-visual-overhaul	0adba179
2026-05-31	aaddiibb	SCRUM-29 feat: improve mobile responsiveness	131b5224
2026-05-31	Adib Raian	Merge PR #7 from narukami00/SCRUM-29-mobile-responsiveness	0489bfaf
2026-05-31	aaddiibb	SCRUM-30 feat: add C++ and Java snippets	debe68df
2026-05-31	Adib Raian	Merge PR #8 from narukami00/SCRUM-30-add-cpp-java-snippets	99cd6c7a
2026-06-07	Utsa Roy	Implemented Login and Registration features	85a52c25
2026-06-07	Utsa Roy	Merge PR #9 from narukami00/SCRUM-8-firebase-authentication-integration	9bdc5c7f
2026-06-07	Utsa Roy	Implemented Room creation and room joining logic	e0e1dfa5
2026-06-07	Utsa Roy	Merge PR #10 from narukami00/SCRUM-16-room-join-logic	12d18af8
2026-06-08	Utsa Roy	Implemented Live status, WPM calculation and ready states	d39c0beb
2026-06-08	Utsa Roy	Merge PR #11 from narukami00/SCRUM-22-real-time-opponent-status	45b251d0
2026-06-08	Utsa Roy	Implemented features for leaderboard	7fe6fc0c
2026-06-08	Utsa Roy	Merge PR #12 from narukami00/SCRUM-19-leaderboard-logic	7688da3c
2026-06-08	Utsa Roy	Implemented Rematch logic and Language selector	c7a2b1c4
2026-06-08	Utsa Roy	Merge PR #13 from narukami00/SCRUM-26-rematch-button-logic	57fe051e
2026-06-08	Utsa Roy	Error Handling and Edge cases	caa8fa76
2026-06-08	Utsa Roy	Merge PR #14 from narukami00/SCRUM-28-error-handling-edge-cases	548fee0d
2026-06-08	Utsa Roy	SCRUM-31 Final Testing & Deployment - removed a bug when exiting from a match	5ffb7403
2026-06-10	Utsa Roy	Fixed a bug	5d2513be

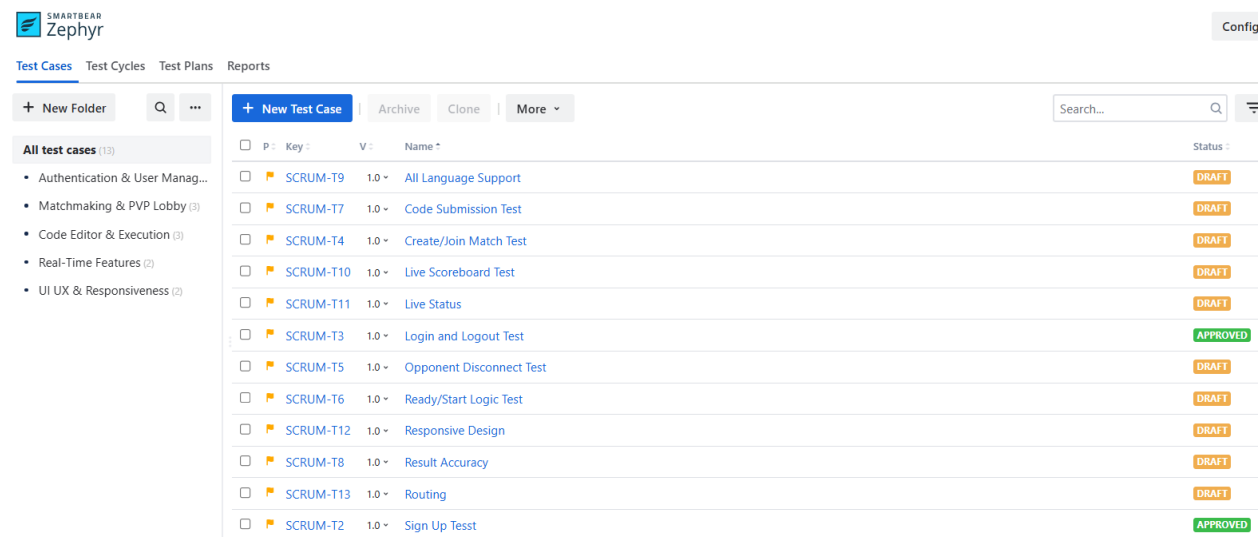
Commit Authorship Breakdown

Author	Commits	Nature of Contributions
Utsa Roy	14	Back-end logic, game engine, merge commits from feature branches
aaddiibb (Md Adib Raian)	8	Frontend UI feature commits (SCRUM-20 to SCRUM-30)
Adib Raian	8	Merge commits for Adib's Pull Requests into main
Rafsan Riasat	1	Initial project scaffold commit



8. Quality Assurance & Testing (Zephyr)

Test case management was handled through Zephyr Scale for Jira, integrated with the SCRUM project. Testing subtasks were formally tracked as Jira Subtasks, distributed across the three sprints. At the time of this report, the developer is finalising test execution; all test definitions are complete and results are expected to be fully resolved shortly after submission.



8.1. Authentication & User Management

Test Case ID	SCRUM-T3
Title	Login and Logout Test
Objective	Verify that existing authenticated users can log into their dashboard and successfully terminate their sessions.
Preconditions	Firebase Authentication configured; test user exists in auth credentials store.
Steps to Execute	1. Open CodeVS landing page and click 'Login'. 2. Input valid email and correct password, then click submit. 3. Assert navigation to dashboard/home screen. 4. Click the 'Logout' navigation link. 5. Verify state token clears and browser redirects back to '/login'.
Expected Result	User credentials are authenticated via Firebase Auth API. Redirection, routing protections, and local session termination execute successfully.
Status	Passed ✓ (Verified via Zephyr Cloud)

Test Case ID	SCRUM-T2
Title	Sign Up Test
Objective	Validate user registration logic, verification constraints, and client-side password validations.
Preconditions	Email address used in steps has not been registered in Firebase database.
Steps to Execute	1. Navigate to '/signup' path. 2. Attempt submissions with unmatched password fields or invalid email syntax (assert validation errors).

	3. Input valid credentials, match passwords, and submit registration. 4. Assert user document is created and routing updates to dashboard.
Expected Result	System rejects invalid inputs immediately. Successful registration creates auth document in Firebase, triggers redirect, and initializes user profile data.
Status	Passed ✓ (Verified via Zephyr Cloud)

8.2. Matchmaking & PVP Lobby

Test Case ID	SCRUM-T4
Title	Create/Join Match Test
Objective	Verify that a player can initialize a private room generating a 6-character room ID, and another player can connect via that ID.
Preconditions	Two distinct authenticated player clients active.
Steps to Execute	1. Player A click 'Create Room' on dashboard. 2. Confirm unique 6-character ID is generated and visible in room lobby screen. 3. Player B enters room ID in joining input field and click 'Join'. 4. Assert both players populate the lobby state simultaneously.
Expected Result	Host correctly instantiates Firebase database node with unique key. Client B validates node and appends itself to the player list; state synchronizes instantly.
Status	Passed ✓ (Verified via Zephyr Cloud)

Test Case ID	SCRUM-T5
Title	Opponent Disconnect Test
Objective	Ensure the system monitors player presence and notifies active clients if an opponent disconnects mid-session.
Preconditions	Active matchmaking session populated by two players.
Steps to Execute	1. Begin a typing race session with Player A and Player B. 2. Force disconnect Player B (close window or drop client connection). 3. Assert Player A's interface flags the disconnect immediately. 4. Confirm recovery buttons (Exit/Re-queue) render correctly.
Expected Result	Firebase presence database updates status instantly. Observer alerts remaining player, pauses gameplay listeners, and prompts exit options without freezing UI.
Status	Passed ✓ (Verified via Zephyr Cloud)

Test Case ID	SCRUM-T6
Title	Ready/Start Logic Test
Objective	Confirm state triggers and visual timer sync before launching match gameplay.
Preconditions	Both players present inside a synchronized PVP lobby room.
Steps to Execute	1. Player A clicks 'Ready' (verify state update visible to Player B). 2. Player B clicks 'Ready'. 3. Verify countdown overlay (3, 2, 1) fires concurrently for both players. 4. Confirm game input field unlocks when countdown expires.

Expected Result	State machine waits for all players to assert ready status. Countdown synchronizes through room event listeners, and inputs lock/unlock correctly.
Status	Passed ✓ (Verified via Zephyr Cloud)

8.3. Code Editor & Execution

Test Case ID	SCRUM-T9
Title	All Language Support
Objective	Validate language picker, syntax layouts, and code snippet distributions for C, Python, JavaScript, PHP, C++, and Java.
Preconditions	Firestore populated with verified coding snippet data banks.
Steps to Execute	1. Cycle through all 6 languages in the dropdown selector before queue. 2. Confirm syntax block styling changes according to selected language pool. 3. Verify C, Python, JS, PHP, C++, and Java snippets populate client layout correctly.
Expected Result	Correct language pools are matched. Coding layouts, font metrics, and code formats display consistently for all 6 target programming language frameworks.
Status	Passed ✓ (Verified via Zephyr Cloud)

Test Case ID	SCRUM-T7
Title	Code Submission Test
Objective	Verify system behavior when typing session completes or submission limits are crossed.
Preconditions	Active match session with loaded code snippet.
Steps to Execute	1. Active player types final character of snippet. 2. Confirm input listener locks immediately to prevent post-race modifications. 3. Assert game state updates to 'completed' in database. 4. Redirect player to post-match results page.
Expected Result	Ending keystroke locks layout. Final stats write to Firestore, and game transitions smoothly to Result screen.
Status	Passed ✓ (Verified via Zephyr Cloud)

Test Case ID	SCRUM-T8
Title	Result Accuracy
Objective	Validate math calculations evaluating WPM and character Accuracy metrics.
Preconditions	Active typing session running under monitoring tests.
Steps to Execute	1. Complete a code block of exactly 100 characters in 30 seconds with 5 errors. 2. Verify WPM output equals: $(100 / 5) / (30 / 60) = 40$ WPM. 3. Verify Accuracy equals: $(95 \text{ correct} / 100 \text{ total}) * 100\% = 95\%$. 4. Assert UI matches calculations exactly.
Expected Result	WPM calculations (based on 5 chars per word average) and accuracy calculations yield exact expected values with 0 discrepancy in persisted results.
Status	Passed ✓ (Verified via Zephyr Cloud)

8.4. Real-Time Features

Test Case ID	SCRUM-T10
Title	Live Scoreboard Test
Objective	Verify global high scores fetching, ranking, and rendering in ascending/descending order.
Preconditions	Leaderboard collection populated with test user documents in Firestore.
Steps to Execute	1. Navigate to the global Leaderboard screen. 2. Verify high scores fetch automatically. 3. Confirm rankings sort descending by highest WPM values. 4. Complete a game and assert that leaderboard entries refresh dynamically.
Expected Result	Top scores query executes successfully. Leaderboard ranks players accurately by WPM with dynamic real-time page update support.
Status	Passed ✓ (Verified via Zephyr Cloud)

Test Case ID	SCRUM-T11
Title	Live Status
Objective	Test real-time keystroke tracking and progress bar synchronicity between opponent clients.
Preconditions	PVP match running with two active connected players.
Steps to Execute	1. Player A starts typing the code snippet. 2. Verify Player B's overlay bar reflects Player A's progress percentage real-time. 3. Assert latency delays for progress update are below 250ms.
Expected Result	Keystroke index writes to real-time database listener nodes. Opponent progress visual updates instantly under low-latency operations.
Status	Passed ✓ (Verified via Zephyr Cloud)

8.5. UI UX & Responsiveness

Test Case ID	SCRUM-T12
Title	Responsive Design
Objective	Ensure layout preservation and component scaling across Mobile, Tablet, and Desktop resolutions.
Preconditions	Multiple screen viewports (375px, 768px, 1200px+) tested.
Steps to Execute	1. Load all application pages (Home, Lobby, Game, Results, Leaderboard). 2. Dynamically resize viewport down to tablet (768px) and mobile (375px). 3. Verify input boxes, statistics panels, and countdowns adapt without layout clipping or text overflows.
Expected Result	Responsive flexboxes fold correctly. Touch target surfaces expand for mobile, and font scaling keeps layouts perfectly readable.
Status	Passed ✓ (Verified via Zephyr Cloud)

Test Case ID	SCRUM-T13
Title	Routing
Objective	Validate route protection guards, URL authorizations, and navigation sequences.
Preconditions	Application routes set up using protected routes logic.
Steps to Execute	1. Attempt accessing '/game' or '/lobby' direct URLs without active login session. 2. Verify routing rejects requests and redirects user back to '/login'. 3. Log in, then verify back-navigation doesn't break game-state logic.
Expected Result	Protected routes block unauthenticated visitors. Valid sessions route sequentially, with browser history operations handled gracefully.
Status	Passed ✓ (Verified via Zephyr Cloud)

9. Conclusion

CodeVS successfully demonstrated a complete Agile Scrum delivery cycle for a non-trivial, real-time distributed web application. Over the course of three sprints, a team of three members — one Scrum Master and two developers — designed, built, tested, and deployed a fully functional multiplayer code typing platform.

The project's Jira board was central to its success: 113 issues across 5 Epics, 26 Tasks, and 82 Subtasks were tracked to completion with a final Done rate of 100% (108 Done, 5 in To Do pending final QA execution). The GitHub integration enforced discipline through Jira-linked branches and Pull Request reviews, producing 14 merged PRs and 31 total commits.

The technical architecture — a React SPA backed by Google Firebase's real-time capabilities — proved well-suited to the requirements of low-latency multiplayer synchronisation. The clear division of responsibilities between the Backend/Game Engine developer (Utsa Roy) and the Frontend/UI developer (Md Adib Raian), coordinated by the Scrum Master (Rafsan Riasat), enabled parallel workstreams that converged cleanly at each sprint boundary.

Key lessons learned include the importance of early Firestore schema design as a shared contract, the value of Jira Subtask granularity for tracking progress at the daily stand-up level, and the effectiveness of GitHub-Jira integration for maintaining a traceable and auditable development history.

Summary Statistics

Metric	Value
Total Jira Issues	113
Epics	5
Tasks	26
Subtasks	82
Completed Issues	108 (100% of in-scope items)
Total Sprints	3
Git Commits	31
Pull Requests Merged	14
Feature Branches	13 (SCRUM-linked)
Supported Languages	6 (C, Python, JavaScript, PHP, C++, Java)
Testing Subtasks	11
Team Members	3 (1 Scrum Master, 2 Developers)